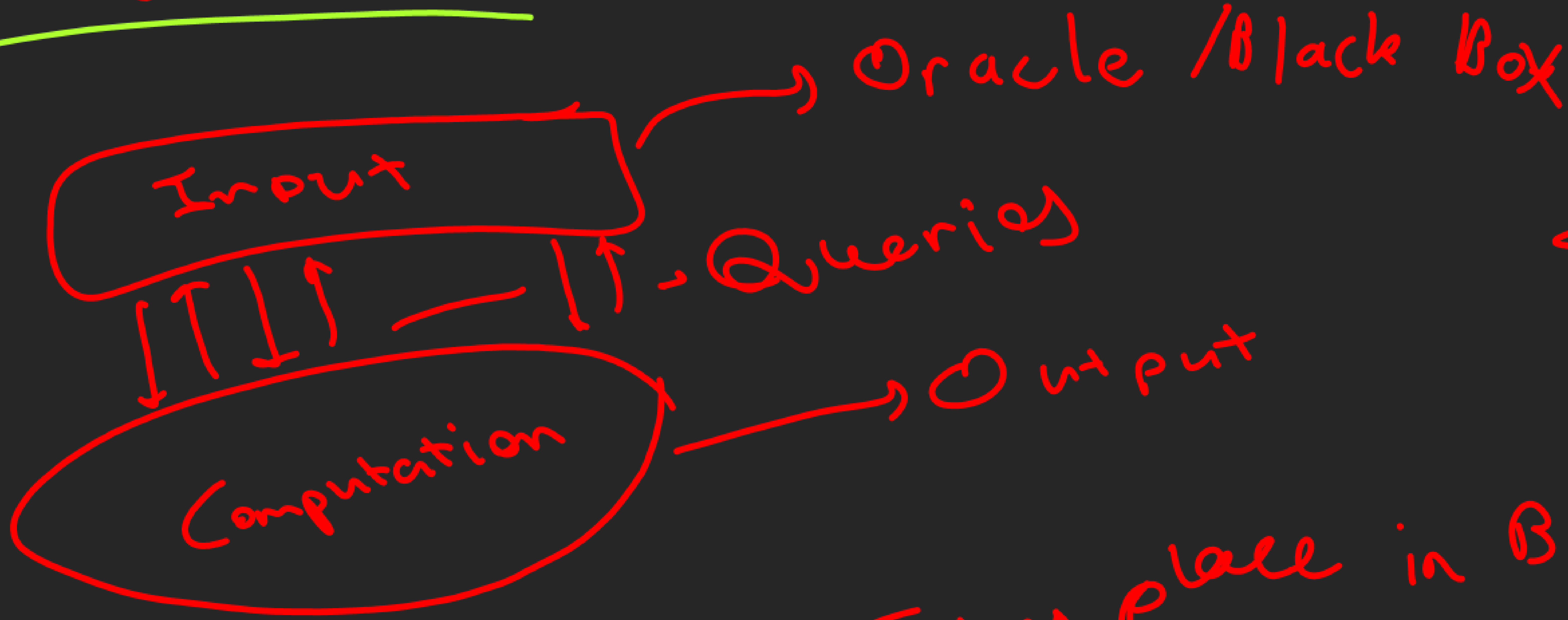
Understanding Quantum Information and Computation

Query model



$\Sigma \in \{0,1\}$

$\Sigma^2 \in \{00,01,10,11\}$

$f:\Sigma^n \to \Sigma^m \to$ Takes place in Black Box

Comp. $\xrightarrow[x\in\Sigma^n]{\text{Query}}$ Black Box $\xrightarrow[f(x)\in\Sigma^m]{\text{made available}}$ Comp.

Examples of Query Problems

→ OR Problem

Input → $f : \Sigma^n \to \Sigma$

Output → 1 if there exists $x \in \Sigma^n$ for which $f(x) = 1$

→ 0 otherwise

→ Parity Problem

Input → $f : \Sigma^n \to \Sigma$

Output → 0 if $f(x) = 1$ for an even no. of strings $x \in \Sigma^n$

→ 1 if $f(x) = 1$ for an odd no. of strings $x \in \Sigma^n$

→ Minimum Problem

Input → $f : \Sigma^n \to \Sigma^m$

Output → The string $y \in \{f(x) : x \in \Sigma^n\}$ that comes first in the lexicographical ordering of Σ^m

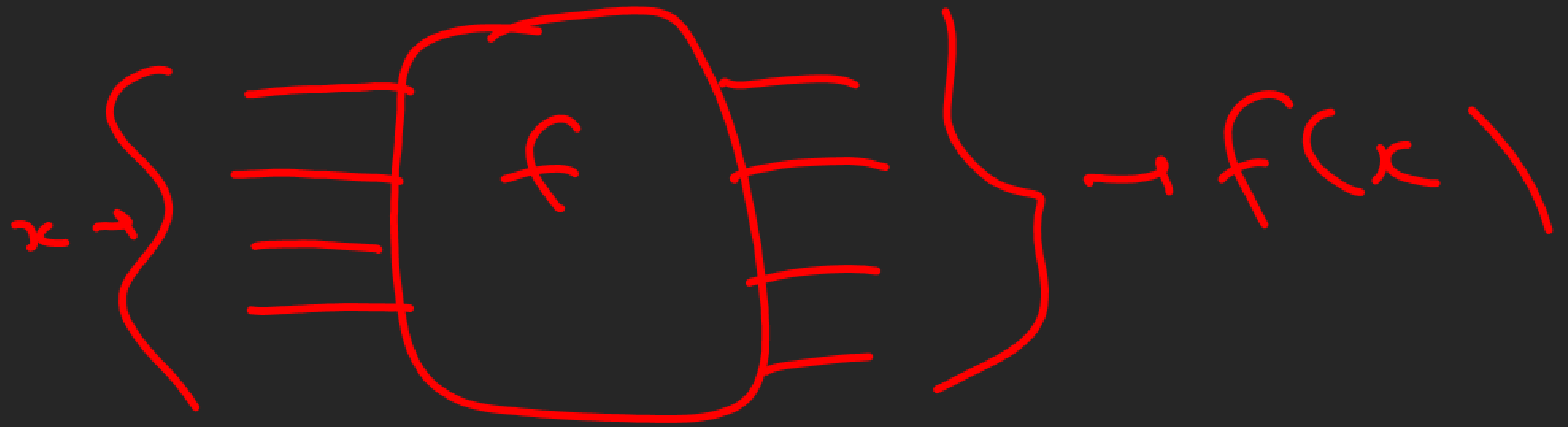
→ Unique Search

Input → $f : \Sigma^n \to \Sigma$

Promise → There is exactly one string $x \in \Sigma^n$ for which $f(x) = 1$ ($f(x) = 0$ for all other strings)

Output → The string x

Query Gates



→ Classical Setup

$$U_f(|x\rangle|y\rangle) = |x\rangle|y \oplus f(x)\rangle$$

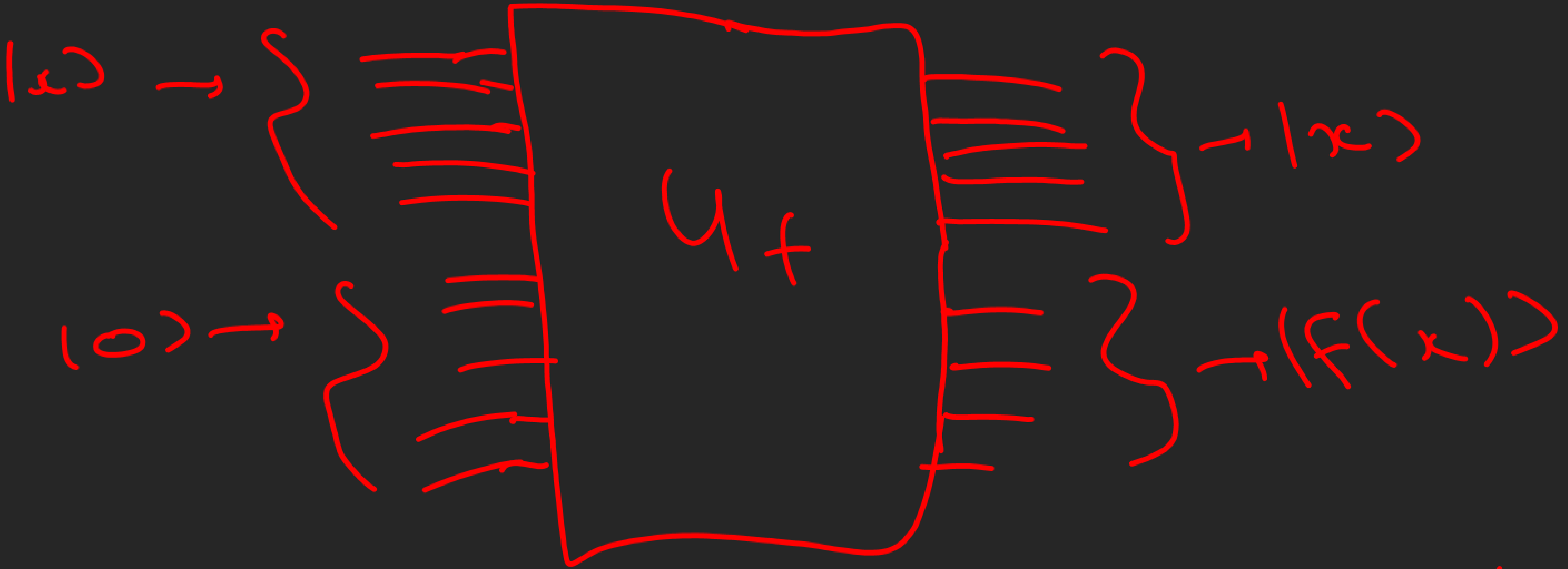
If $y=0 \rightarrow U_f(|x\rangle|0\rangle) = |x\rangle|f(x)\rangle$

$U_f \rightarrow$ Query Gate (unitary)

$x \in \Sigma^n$

$y \in \Sigma^m$

$f: \Sigma^n \rightarrow \Sigma^m$



Grover's Algorithm $\rightarrow \langle s|x\rangle|00\rangle|0\rangle = |x\rangle|00\rangle|f(x)\rangle$

@quantum country

$U_f^{-1} = U_f \quad U_f^{\dagger} = U_f$